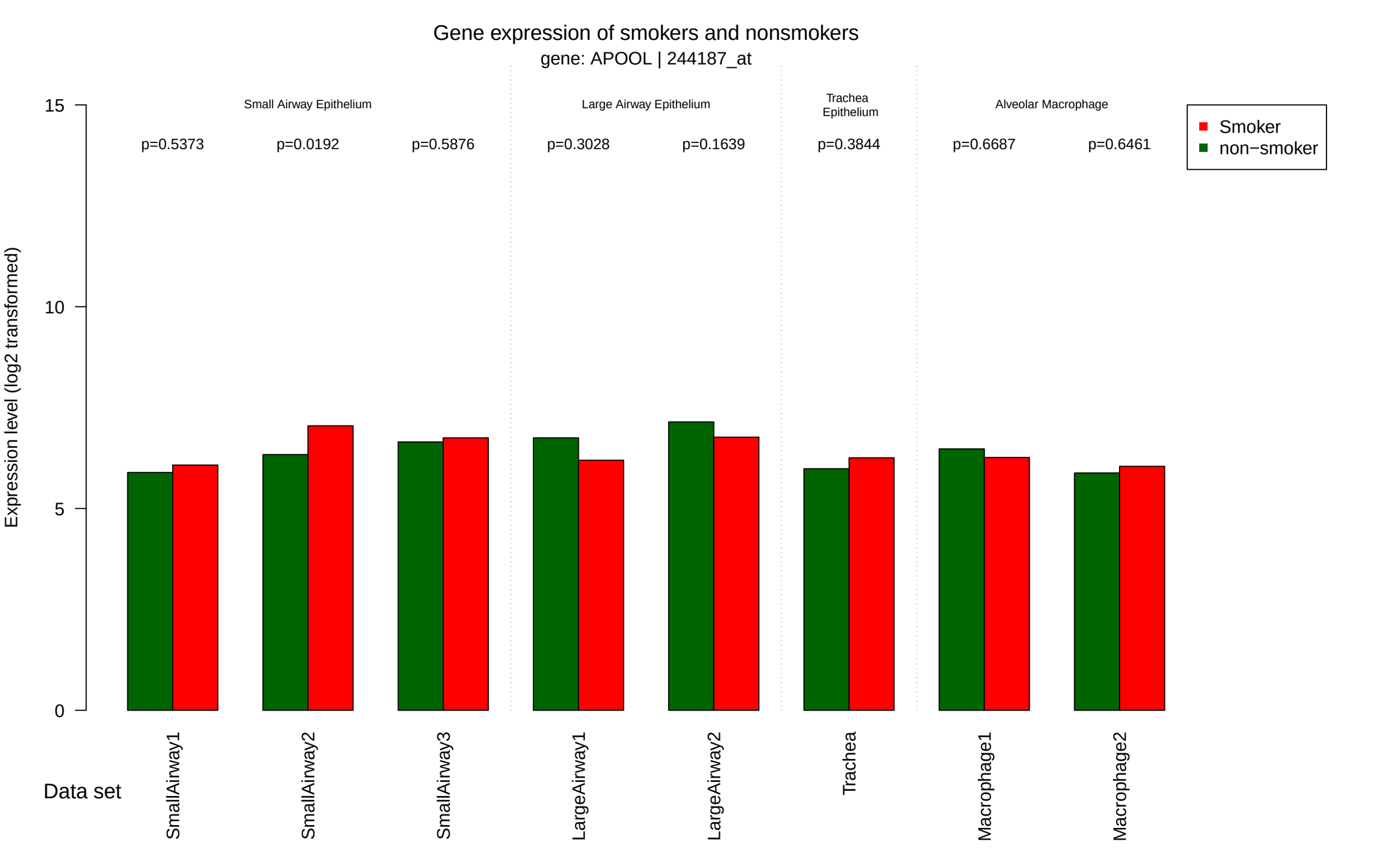
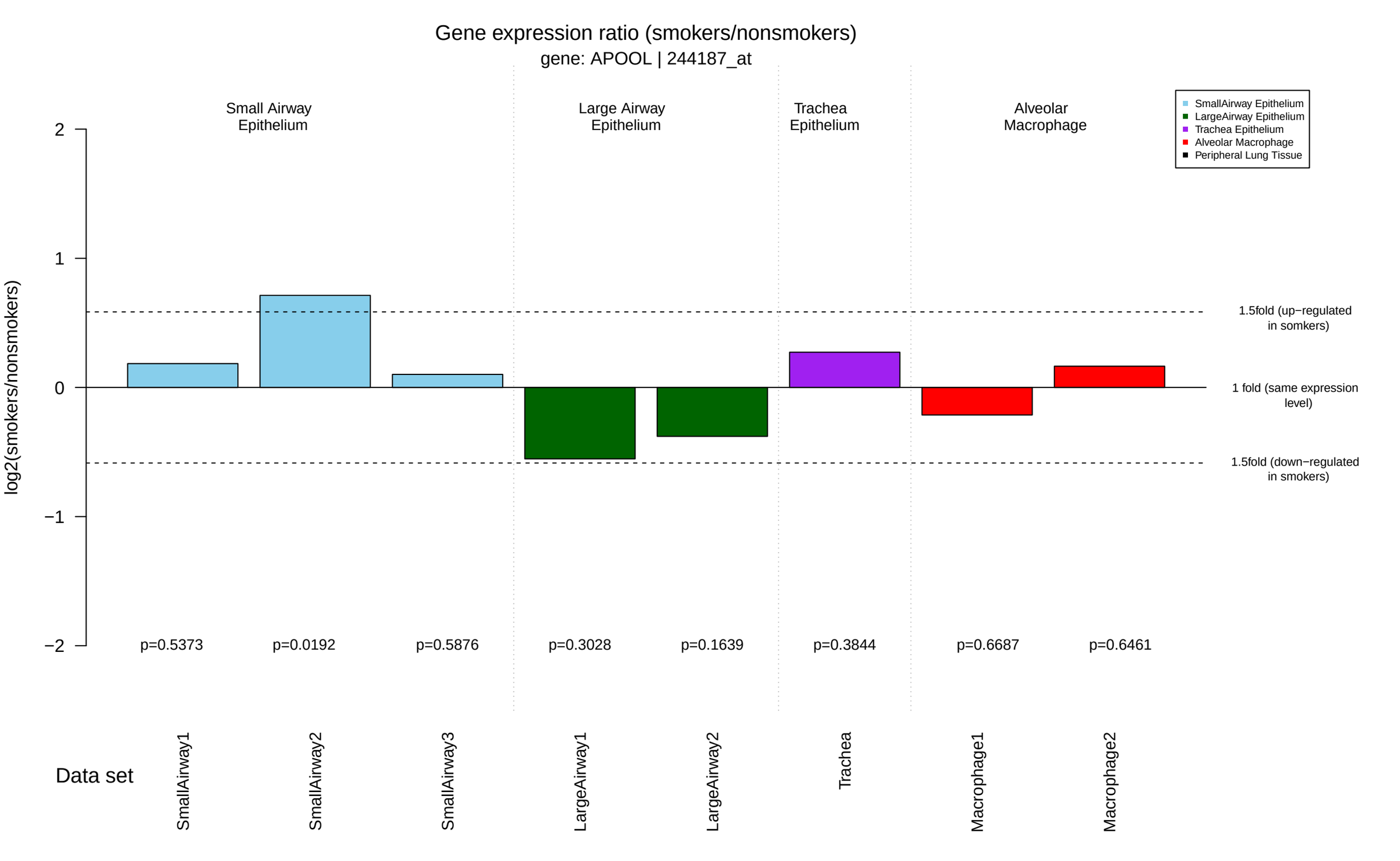
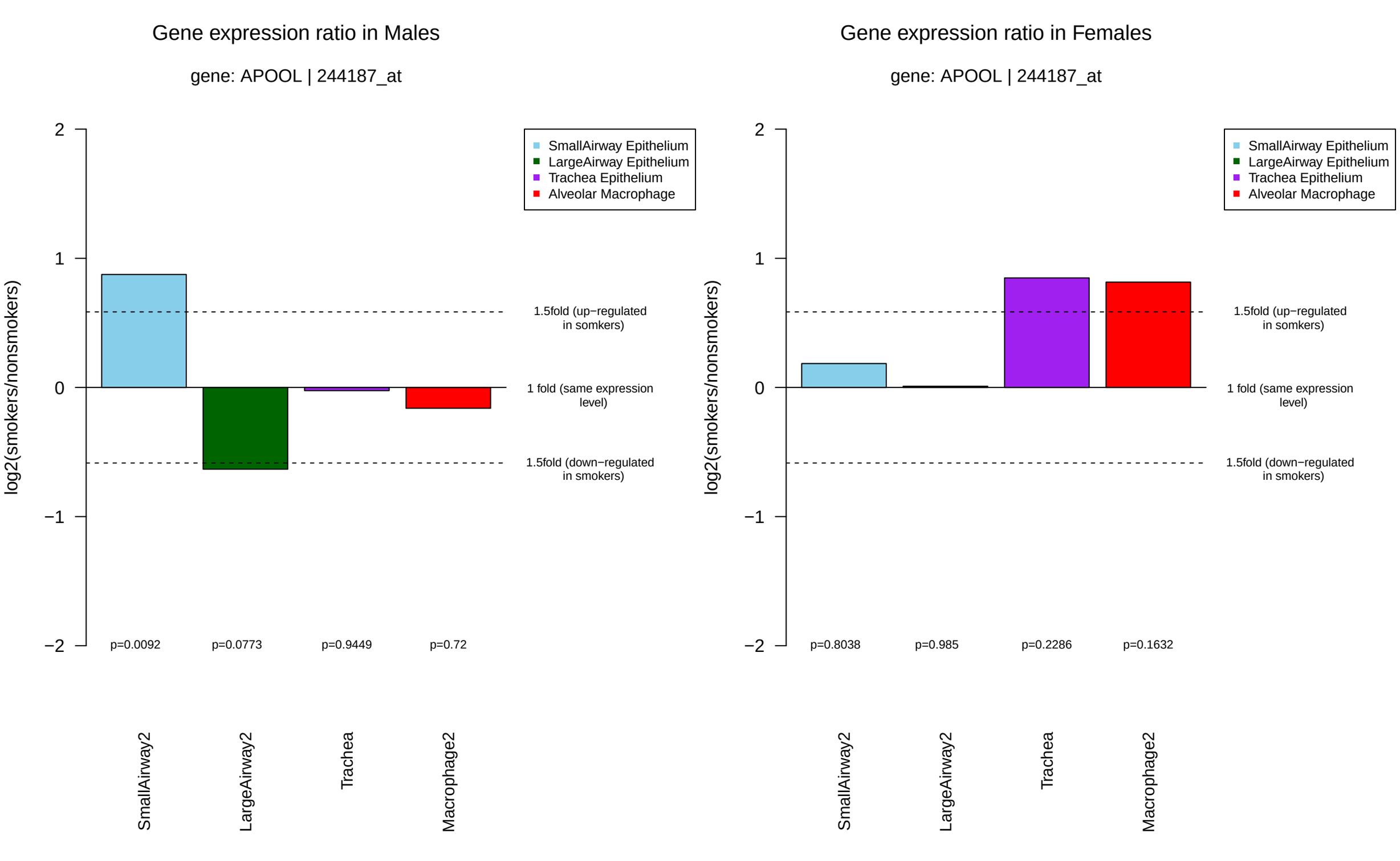
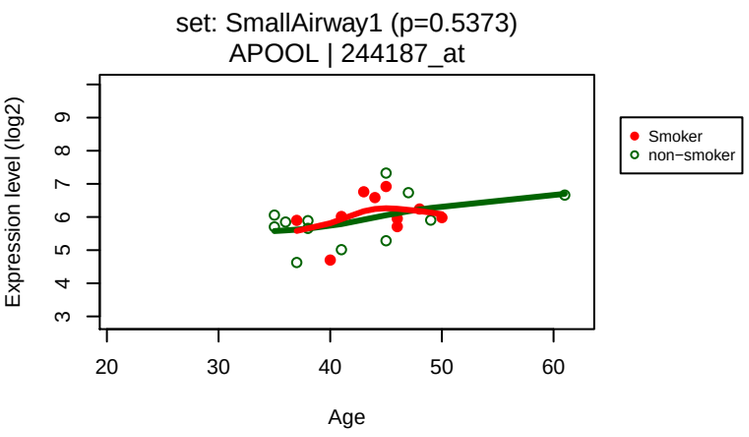
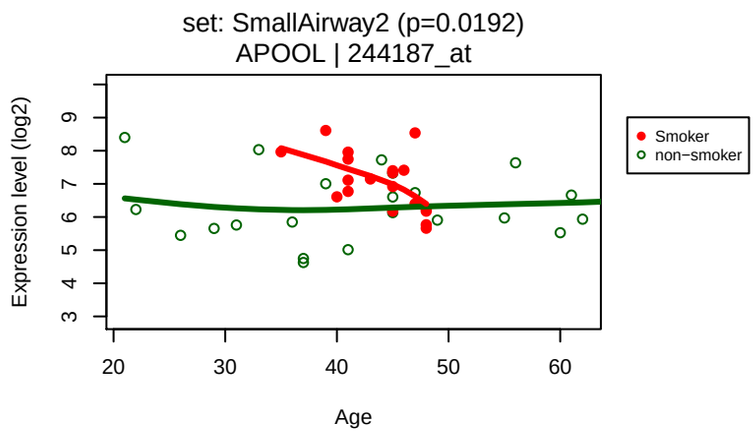
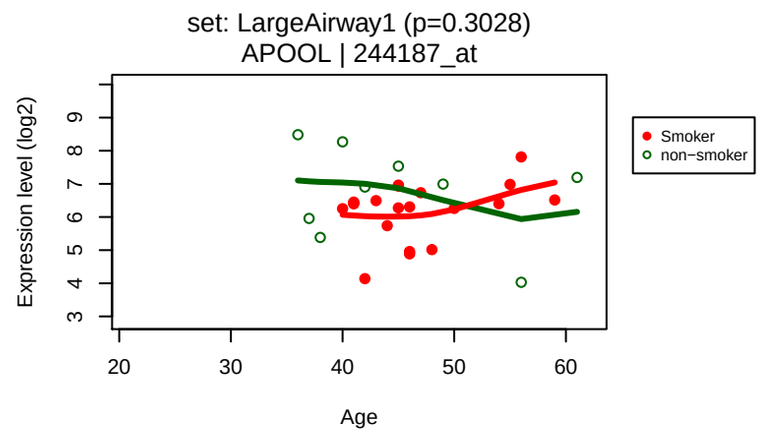

Gene expression of smokers and nonsmokers
gene: APOOL | 244187_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5373
p=0.0192
p=0.5876
p=0.3028
p=0.1639
p=0.3844
p=0.6687
p=0.6461
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

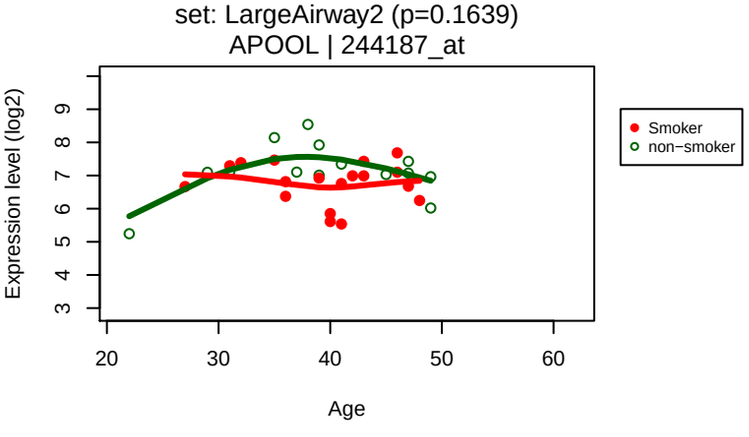

Gene expression ratio (smokers/nonsmokers)
gene: APOOL | 244187_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5373
p=0.0192
p=0.5876
p=0.3028
p=0.1639
p=0.3844
p=0.6687
p=0.6461
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

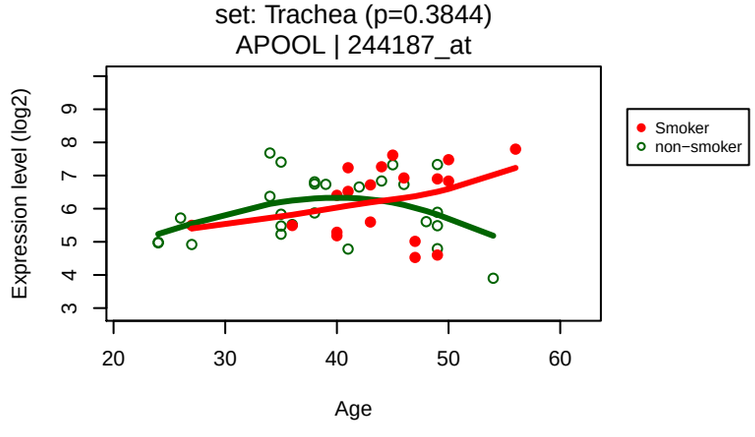

Gene expression ratio in Males
gene: APOOL | 244187_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0092
p=0.0773
p=0.9449
p=0.72
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APOOL | 244187_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8038
p=0.985
p=0.2286
p=0.1632
SmallAirway2
LargeAirway2
Trachea
Macrophage2

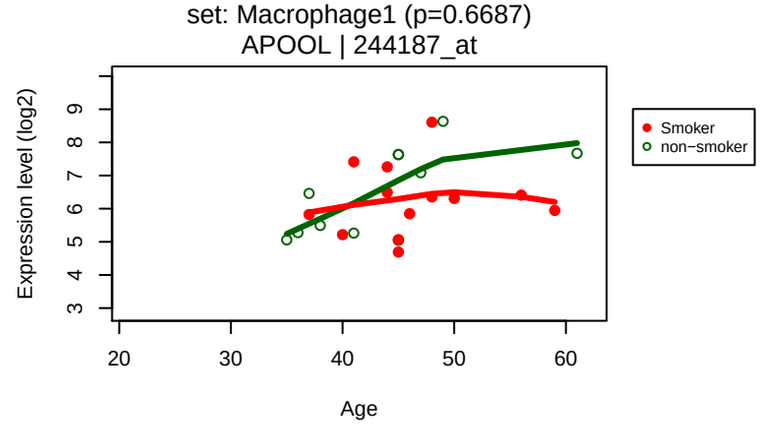

set: SmallAirway1 (p=0.5373)
APOOL | 244187_at
Expression level (log2)
Age
Smoker
non−smoker

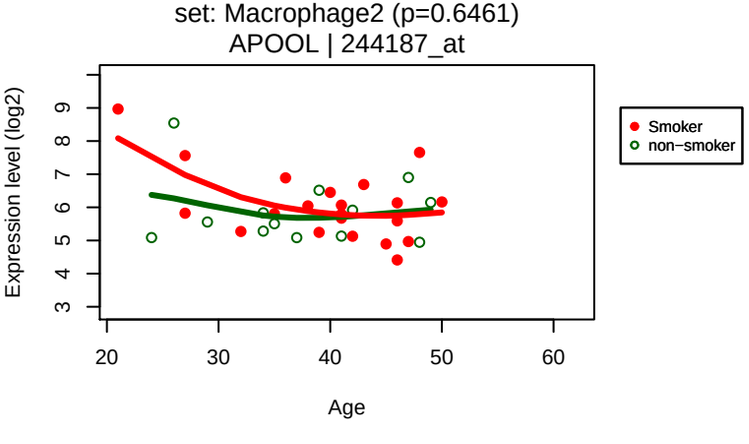

set: SmallAirway2 (p=0.0192)
APOOL | 244187_at
Expression level (log2)
Age
Smoker
non−smoker

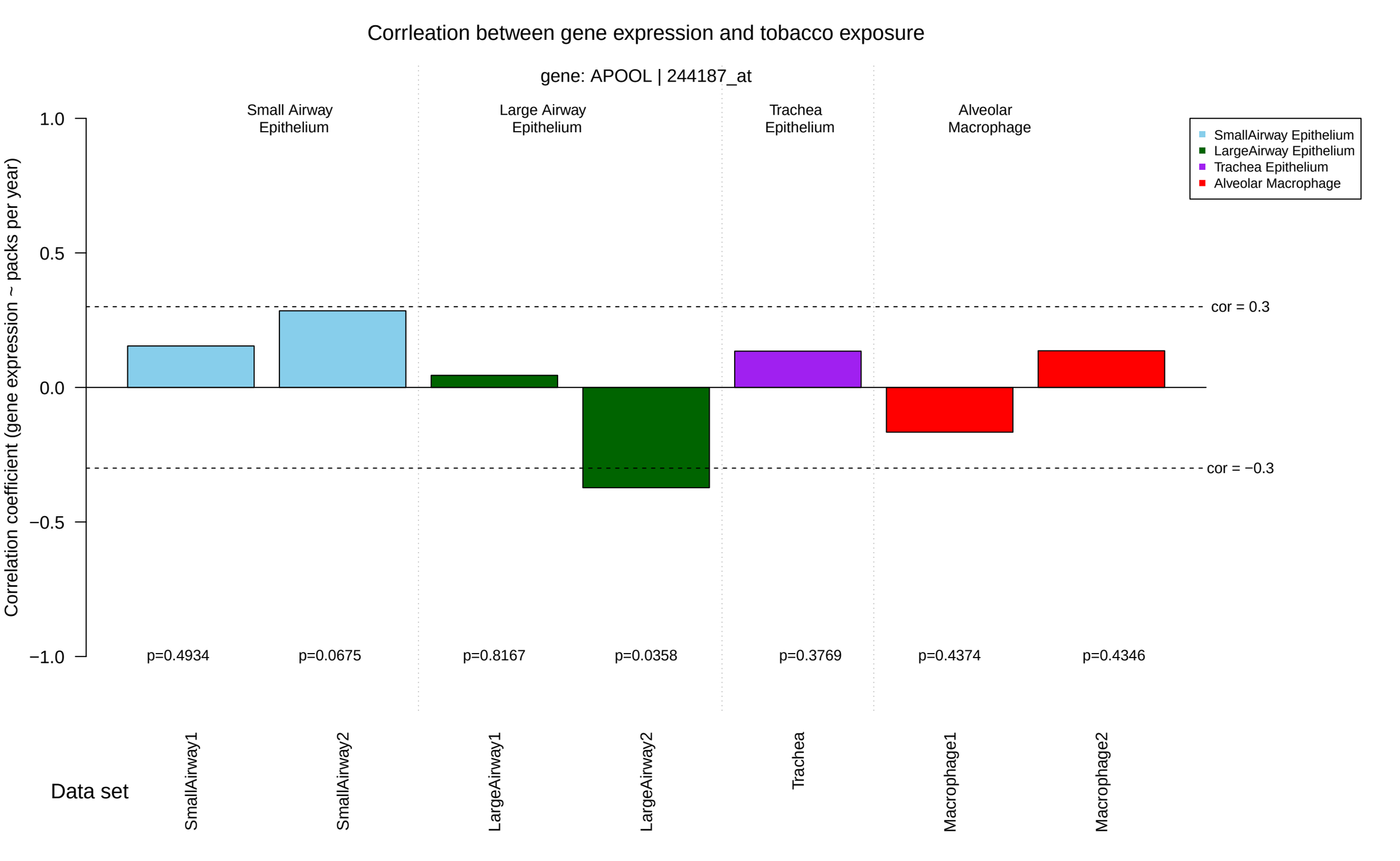

set: LargeAirway1 (p=0.3028)
APOOL | 244187_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1639)
APOOL | 244187_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3844)
APOOL | 244187_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6687)
APOOL | 244187_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6461)
APOOL | 244187_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APOOL | 244187_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4934
p=0.0675
p=0.8167
p=0.0358
p=0.3769
p=0.4374
p=0.4346
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2